

Mayuravel Pakkiarajah

Software Engineer | Full-Stack Developer

📍 Batticaloa, Sri Lanka | ✉ mayuravelp@gmail.com | ☎ +94 750683305 | in LinkedIn | 🐙 GitHub | 🌐 Portfolio

Profile

Software Engineer and Information Technology graduate with hands-on experience developing full-stack applications using Java, Spring Boot, Angular, C, and ASP.NET Core. Experienced in designing RESTful APIs, microservices-based applications, database systems, and scalable software solutions. Strong foundation in software development, problem-solving, and modern development practices, with additional experience in Python and computer vision.

Experience

Software Engineer Intern - NAVITSA TECHNOLOGIES (PVT) LTD. (Mar 2025 - Sep 2025)

1. ERP System

- Developed a full-stack ERP solution for supermarket inventory and sales management using ASP.NET Core and MSSQL.
- Implemented MVC architecture, business logic, data validation, and database schema optimization.

2. LEDGER CHIMP

- Developed responsive web interfaces using Angular.
- Built backend services using Spring Boot and MySQL within a microservices architecture.

3. Banking System

- Developed responsive Angular interfaces and RESTful APIs using Spring Boot for a loan-management system.
- Applied microservices architecture to support independent deployment and scalability.

Skills

Languages: Java, Go, C#, TypeScript, JavaScript, Python, C, C++

Frameworks & Libraries: Spring Boot, Spring Cloud, Angular, ASP.NET Core

AI/ML: PyTorch, YOLOv8, OpenCV, NumPy

Databases & Tools: MySQL, MSSQL, RabbitMQ, Docker, Git, GitHub, Figma, Postman

Education

University of Moratuwa - Faculty of Information Technology

2022 - 2026 (Expected)

BSc. (Hons.) in Information Technology

Projects

Machine Learning Approaches for Dashcam-Based Road Safety Monitoring 🚗

- Dashcams are mainly used for post-incident evidence rather than real-time assistance. Conventional vehicles, especially in developing countries, lack affordable integrated systems that provide timely warnings for road signs, road damage, lane departure, and critical road anomalies, increasing accident risk.
- I developed a lane-line detection and lane-departure warning module using YOLOv8s-seg to identify lane boundaries and road markings from dashcam footage.
- Fine-tuned and evaluated the segmentation model, achieving 90.8% precision, 91.4% recall, 92.9% mAP@0.5, and 76.4% mAP@0.5:0.95.
- Implemented lane-boundary estimation and lateral-offset calculation to determine vehicle position relative to the detected lane.
- Developed temporally stable lane-departure detection and left/right warning logic for dashcam video.
- Technologies: Python, PyTorch, YOLOv8-seg, OpenCV, NumPy, Kaggle, Docker

Online Learning Platform 🎓

- Developing a full-stack online learning platform supporting course creation, enrollment, quizzes, progress tracking, certificates, reviews, notifications, and analytics
- Designing a microservices-based backend using Spring Boot, Spring Cloud, API Gateway, Eureka, and MySQL.
- Developing role-based Angular interfaces for Students, Instructors, and Administrators.
- Integrating RabbitMQ for asynchronous communication and Docker for containerized deployment.
- Technologies : Angular, Spring Boot, Spring Cloud, MySQL, RabbitMQ, Docker, Bootstrap, Micro Service Architecture

Weekly Report Generator & Team Dashboard

- Teams tracking weekly progress across projects typically rely on ad-hoc spreadsheets or chat updates, making it hard for managers to enforce a consistent reporting format, track correction cycles, or get a consolidated view of team-wide status, blockers, and workload.
- I designed and built a full-stack reporting platform with role-based access for team members and managers, enforcing a fixed report structure so every submission is directly comparable across the team.
- Implemented a submit-review-correction workflow with a state machine where every edit after a correction request creates a new immutable report version rather than overwriting content, preserving full review history.
- Enforced role-based access control at both the API and service layer using Spring Security and JWT, ensuring team members cannot access another member's data even by ID manipulation; verified this with an automated RBAC test suite.
- Enforced role-based access control at both the API and service layer using Spring Security and JWT, ensuring team members cannot access another member's data even by ID manipulation; verified this with an automated RBAC test suite.
- Built a manager analytics dashboard with live charts computed via aggregate JPA queries, plus filterable/paginated report search across the whole team.
- Integrated an AI chat assistant using OpenAI function calling, giving managers natural-language Q&A over team activity and AI-generated weekly summaries, grounded in real report data via scoped, read-only tool calls rather than a fixed prompt context.
- Technologies: Angular, Spring Boot, Spring Security, MySQL, JWT, OpenAI API, Chart.js, Flyway

Book Network

- Book Network is a full-stack app for managing book collections and engaging with a community of book enthusiasts.
- Technologies : Spring Boot, MySql

Field Work Management System

- The aim of this project is to create a system that links Field Workers and consumers in the restructuring sectors
- Contribution : Developed UI designs using Figma, Built chat module using .NET SignalR, Developed web frontend pages
- Technologies : Angular, ASP.NET Core, MSSQL, Figma

Leadership & Achievements

- President - Tamil Literary Association, University of Moratuwa (2025/2026)
- Batch Representative in 1st year, Faculty of Information Technology, University of Moratuwa (Batch 21)
- Representing University volleyball team - SLUG - 2023, Inter University Meet 2024, SLUG - 2025
- Complete ICTT NVQ - 4 (NAITA - 2018/19)

Reference

Dr. Supunmali Ahangama
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